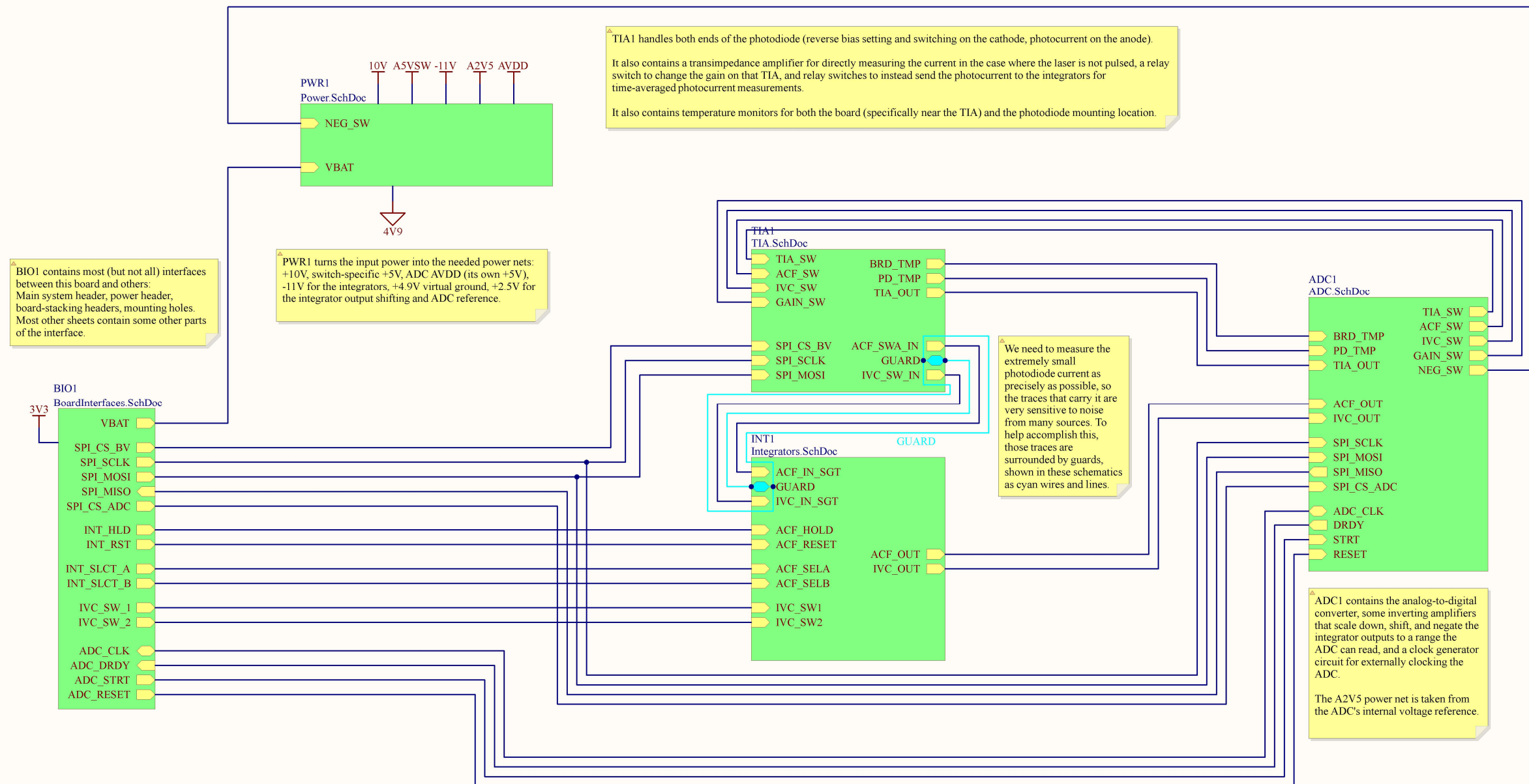


Changes from V26B include:

- Added T1A isolation relays and associated control lines to prevent amplifier contention
- Added 2:4 Decoder to allow for one extra IO relay control and enforce amplifier mutual exclusivity
- Rework power-on sequencing to ensure that ADC AVDD is on before input voltages on its lines rise above allowable thresholds
- Add input anti-aliasing filters for all critical inputs (also provides over/undervoltage protection)
- Fix thermistors lacking any bias
- Updated solder to bring it in-line with V26B baseline
- Various little fixes
 - Fix fast-start for 10V
 - 5V adjustment resistors removed (fixed voltage model)
 - Drop T1A max voltage from 10V to 5V to prevent overvoltage events in case of negative photocurrent

**THIS REVISION, V26_, WAS
SENT FOR FABRICATION ON**

Compatible board:
Soldier: _____



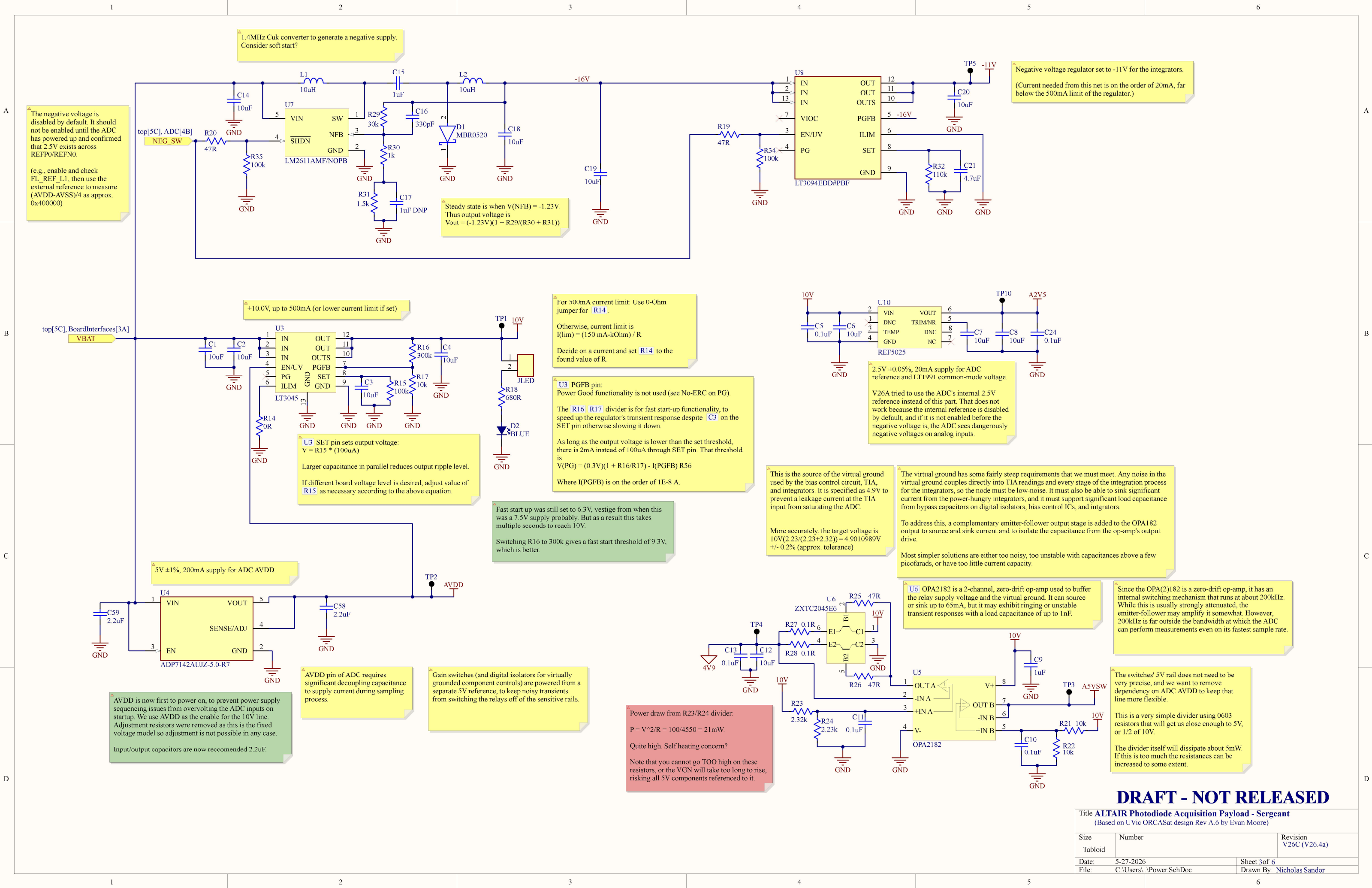
INT1 contains two integrator circuits.

One works with the ACF2101 dual-channel switched integrator, which is the primary measurement device for photocurrents from both the Sergeant board and the stacked Soldier board, connected through spring-loaded pogo pins and pads.

The other works with the IVC102 single-channel switched integrator, which serves to validate the values given by the ACF2101. The Soldier board has its own IVC102 as well.

DRAFT - NOT RELEASED

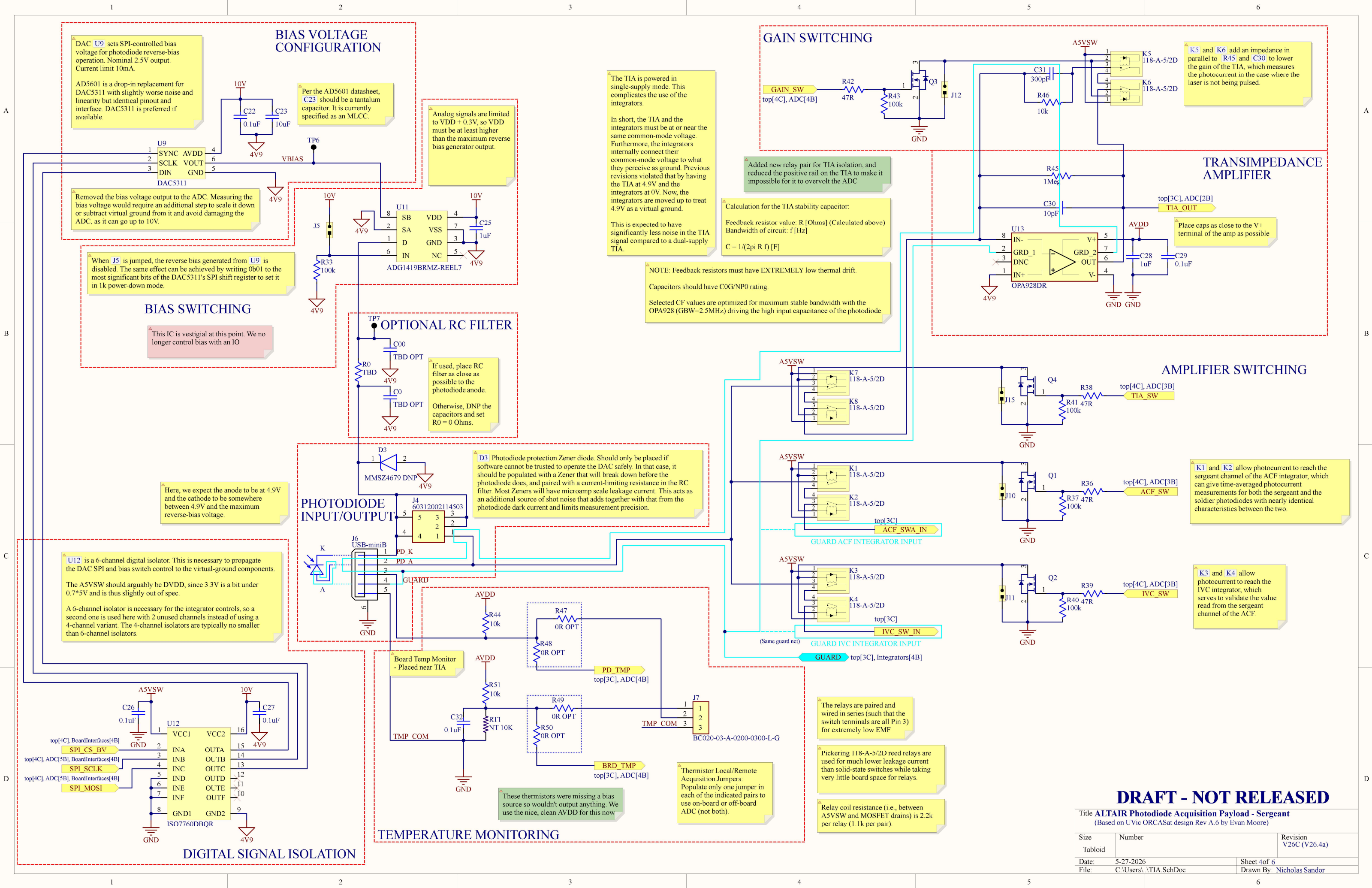
Title ALTAIR Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)			
Size Tabloid	Number	Revision V26C (V26.4a)	
Date:	5-27-2026	Sheet 1 of 6	
File:	C:\Users\j\top_SchDoc	Drawn By: Nicholas Sandor	



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Title **ALTAIR Photodiode Acquisition Payload - Sergeant**
(Based on Uvic ORCASat design Rev A.6 by Evan Moore)

Size	Number	Revision
Tabloid		V26C (V26.4a)
Date:	5-27-2026	Sheet 3 of 6
File:	C:\Users\...\Power.SchDoc	Drawn By: Nicholas Sandor



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Title ALTAIR Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)		
Size	Number	Revision
Tabloid		V26C (V26.4a)
Date:	5-27-2026	Sheet 4 of 6
File:	C:\Users\... \TIA.SchDoc	Drawn By: Nicholas Sandor

A

B

C

D

A

B

C

D

A previous board revision connected ACF COM and SW-COM pins and IVC AGND pin to the respective guards. This was undone on the board because it caused a short circuit. It is now recognized that this was likely because the guard was pulled to 4.9V by the TIA. These are not re-connected here, but it is something to consider restoring with this design.

Sergeant has a pogo pin connector (J8) to Soldier's pads for the output of the ACF integrator.

Soldier has 3 1x2 pogo pin connectors to Sergeant's pads (J9) for the input and guard on its side of the ACF integrator.

R54 R57 R61

These jumpers exist to control where the SW COM pins connect to ground. The routing of the wires on the schematic reflects the desired routing of the traces on the PCB, where those nets should be routed between the two boards' guards to minimize crosstalk between them.

R52 R55 R53 R56

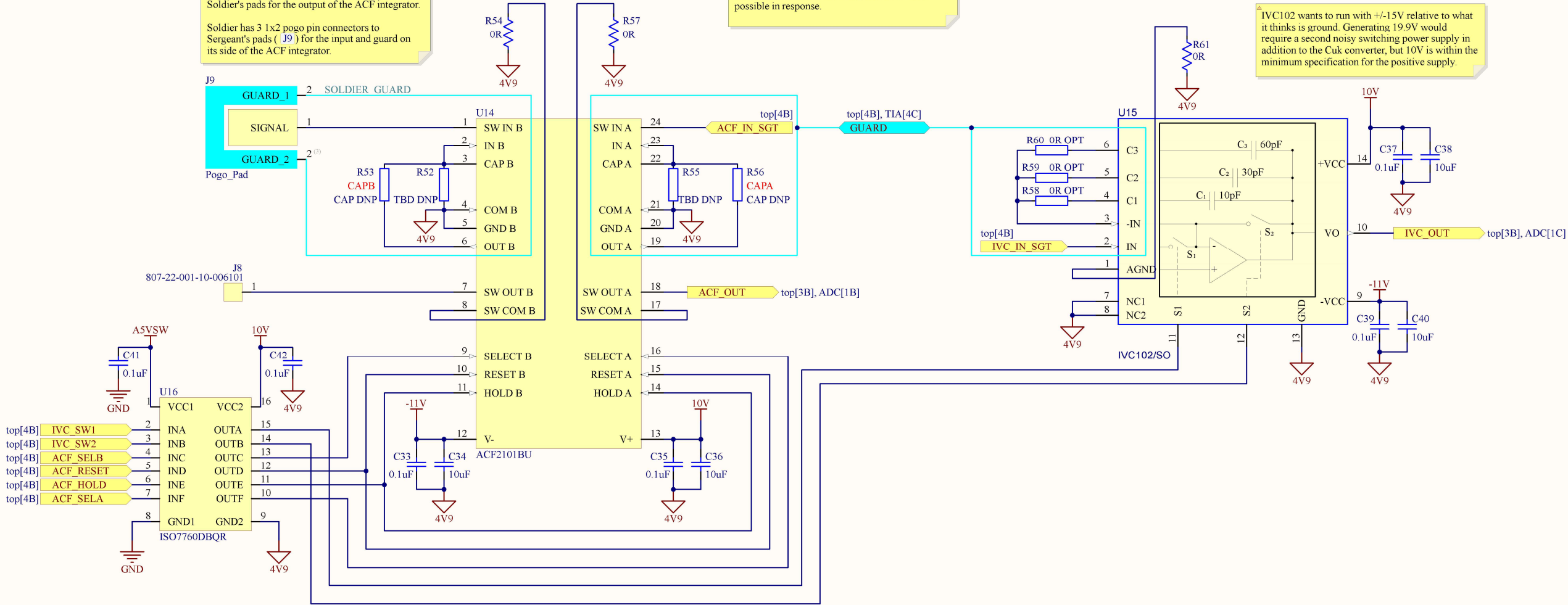
These are nominally DNP. They exist to provide points where we can add capacitors (typically not resistors) to adjust the integrator responses as desired after the board is already fabricated.

However, in doing so, great care must be taken to choose matching pairs of passive components to keep the two sides of the integrator as close as possible in response.

R60 R59 R58

These can be jumpers or capacitors used to fine tune the IVC's response. At least one must be populated to close the op-amp feedback loop.

IVC102 wants to run with +/-15V relative to what it thinks is ground. Generating 19.9V would require a second noisy switching power supply in addition to the Cuk converter, but 10V is within the minimum specification for the positive supply.



U16 is a 6-channel digital isolator. This is necessary to propagate the integrator switch controls to the virtual-ground integrators.

The A5VSW should arguably be DVDD, since 3.3V is a bit under 0.7*5V and is thus slightly out of spec.

OPERATING THE ACF2101		
FUNCTION	HOLD SWITCH	RESET SWITCH
INTEGRATE	CLOSED	OPEN
HOLD	OPEN	OPEN
RESET	OPEN	CLOSED

OPERATING THE IVC102		
FUNCTION	SWITCH 1	SWITCH 2
INTEGRATE	CLOSED	OPEN
HOLD	OPEN	OPEN
RESET	OPEN	CLOSED

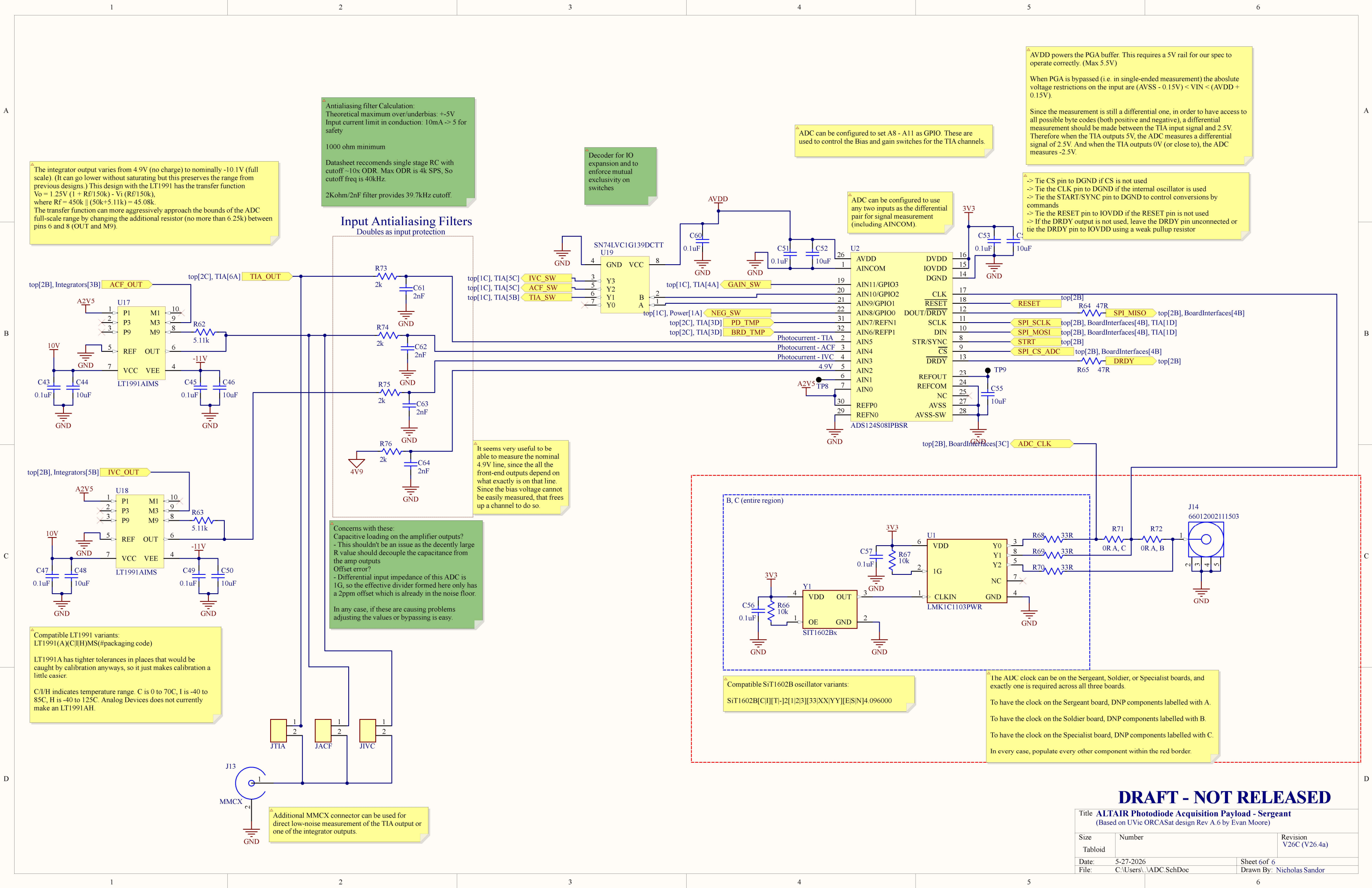
Switches are TTL/CMOS compatible.

Switch Open: Logic HIGH (3.3V or 5.0V)

Switch Closed: Logic LOW (0V)

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The integrator output varies from 4.9V (no charge) to nominally -10.1V (full scale). (It can go lower without saturating but this preserves the range from previous designs.) This design with the LT1991 has the transfer function $V_o = 1.25V (1 + R_f/150k) - V_i (R_f/150k)$, where $R_f = 450k \parallel (50k + 5.11k) = 45.08k$. The transfer function can more aggressively approach the bounds of the ADC full-scale range by changing the additional resistor (no more than 6.25k) between pins 6 and 8 (OUT and M9).

Antialiasing filter Calculation:
Theoretical maximum over/underbias: +5V
Input current limit in conduction: 10mA -> 5 for safety

1000 ohm minimum

Datasheet recommends single stage RC with cutoff ~10x ODR. Max ODR is 4k SPS, So cutoff freq is 40kHz.

2Kohm/2nF filter provides 39.7kHz cutoff.

Decoder for IO expansion and to enforce mutual exclusivity on switches

ADC can be configured to set A8 - A11 as GPIO. These are used to control the Bias and gain switches for the TIA channels.

AVDD powers the PGA buffer. This requires a 5V rail for our spec to operate correctly. (Max 5.5V)

When PGA is bypassed (i.e. in single-ended measurement) the absolute voltage restrictions on the input are $(AVSS - 0.15V) < VIN < (AVDD + 0.15V)$.

Since the measurement is still a differential one, in order to have access to all possible byte codes (both positive and negative), a differential measurement should be made between the TIA input signal and 2.5V. Therefore when the TIA outputs 5V, the ADC measures a differential signal of 2.5V. And when the TIA outputs 0V (or close to), the ADC measures -2.5V.

-> Tie CS pin to DGND if CS is not used
-> Tie the CLK pin to DGND if the internal oscillator is used
-> Tie the START/SYNC pin to DGND to control conversions by commands
-> Tie the RESET pin to IOVDD if the RESET pin is not used
-> If the DRDY output is not used, leave the DRDY pin unconnected or tie the DRDY pin to IOVDD using a weak pullup resistor

ADC can be configured to use any two inputs as the differential pair for signal measurement (including AINCOM).

It seems very useful to be able to measure the nominal 4.9V line, since the all the front-end outputs depend on what exactly is on that line. Since the bias voltage cannot be easily measured, that frees up a channel to do so.

Concerns with these:
Capacitive loading on the amplifier outputs?
- This shouldn't be an issue as the decently large R value should decouple the capacitance from the amp outputs
Offset error?
- Differential input impedance of this ADC is 1G, so the effective divider formed here only has a 2ppm offset which is already in the noise floor.

In any case, if these are causing problems adjusting the values or bypassing is easy.

Compatible LT1991 variants:
LT1991(A)(C)(I)(H)MS(#packaging code)

LT1991A has tighter tolerances in places that would be caught by calibration anyways, so it just makes calibration a little easier.

C/I/H indicates temperature range. C is 0 to 70C, I is -40 to 85C, H is -40 to 125C. Analog Devices does not currently make an LT1991AH.

Additional MMCX connector can be used for direct low-noise measurement of the TIA output or one of the integrator outputs.

B, C (entire region)

Compatible SiT1602B oscillator variants:
SiT1602B[C](I)[T]-[2][1][2][3][33][XX][YY][E][S][N]4.096000

The ADC clock can be on the Sergeant, Soldier, or Specialist boards, and exactly one is required across all three boards.

To have the clock on the Sergeant board, DNP components labelled with A.

To have the clock on the Soldier board, DNP components labelled with B.

To have the clock on the Specialist board, DNP components labelled with C.

In every case, populate every other component within the red border.

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Title ALTair Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)		
Size Tabloid	Number	Revision V26C (V26.4a)
Date: File:	5-27-2026 C:\Users\...\ADC.SchDoc	Sheet 6 of 6 Drawn By: Nicholas Sandor